

124. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4

124.1 Overview

This clause specifies the 400GBASE-DR4 PMD together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 124–1, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Table 124–1—Physical Layer clauses associated with the 400GBASE-DR4 PMD

Associated clause	400GBASE-DR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—PCS	Required
120—PMA	Required
120B—Chip-to-chip 400GAUI-16	Optional
120C—Chip-to-module 400GAUI-16	Optional
120D—Chip-to-chip 400GAUI-8	Optional
120E—Chip-to-module 400GAUI-8	Optional
78—Energy Efficient Ethernet	Optional

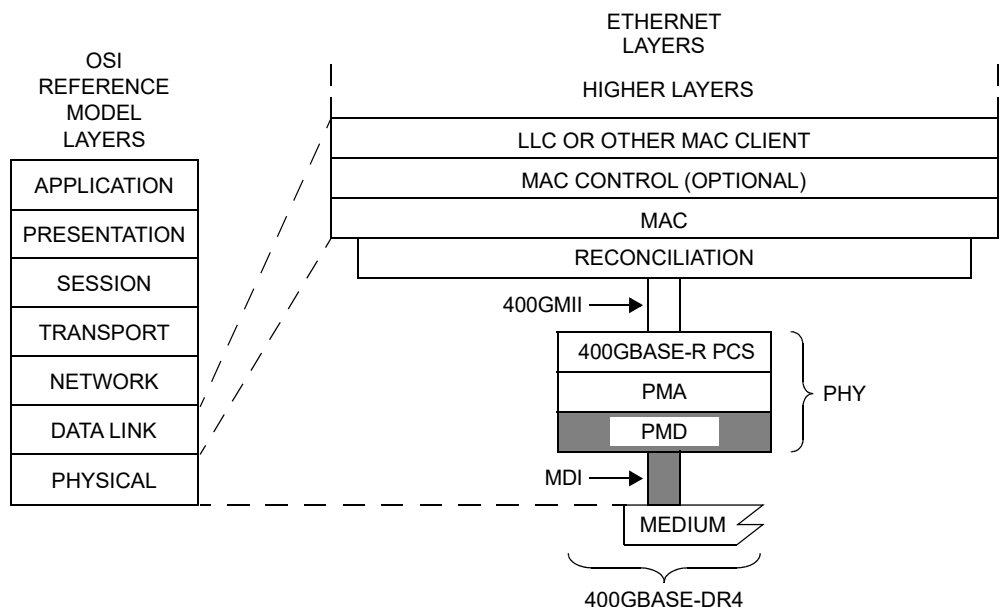
^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

Figure 124–1 shows the relationship of the PMD and MDI (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 400 Gb/s Ethernet is introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2.

400GBASE-DR4 PHYs with the optional Energy Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

124.1.1 Bit error ratio

The bit error ratio (BER) when processed according to Clause 120 shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap when processed according to Clause 120 and then Clause 119. For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces.



400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
 LLC = LOGICAL LINK CONTROL
 MAC = MEDIA ACCESS CONTROL
 MDI = MEDIUM DEPENDENT INTERFACE
 PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
 PMA = PHYSICAL MEDIUM ATTACHMENT
 PMD = PHYSICAL MEDIUM DEPENDENT
 DR = PMD FOR SINGLE-MODE FIBER — 500 m

Figure 124–1—400GBASE-DR4 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.

124.2 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 400GBASE-DR4 PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3. The PMD service interface primitives are summarized as follows:

PMD:IS_UNITDATA_ *i*.request
 PMD:IS_UNITDATA_ *i*.indication
 PMD:IS_SIGNAL.indication

The 400GBASE-DR4 PMD has four parallel symbol streams, hence $i = 0$ to 3.

In the transmit direction, the PMA continuously sends four parallel symbol streams to the PMD, one per lane, each at a nominal signaling rate of 53.125 GBd. The PMD then converts these streams of data units into the appropriate signals on the MDI.

In the receive direction, the PMD continuously sends four parallel symbol streams to the PMA corresponding to the signals received from the MDI, one per lane, each at a nominal signaling rate of 53.125 GBd. See NOTE at the end of 120.5.2 concerning the transition density of lanes operating at this nominal signaling rate.

The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the PMD:IS_SIGNAL.indication(SIGNAL_OK) inter-sublayer service interface primitive defined in 116.3.

The SIGNAL_DETECT parameter can take on one of two values: OK or FAIL. When SIGNAL_DETECT = FAIL, the rx_symbol parameters are undefined.

NOTE—SIGNAL_DETECT = OK does not guarantee that the rx_symbol parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_DETECT = OK indication and still not meet the BER defined in 124.1.1.

124.3 Delay and Skew

124.3.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR4 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 and its references.

124.3.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS. Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116-4 and Figure 116-5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5. The measurements of Skew and Skew Variation are defined in 86.8.3.1.

124.4 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that may provide control and status information for and about the PMD. If the MDIO interface is implemented, the mapping of MDIO

control variables to PMD control variables shall be as shown in Table 124–2 and the mapping of MDIO status variables to PMD status variables shall be as shown in Table 124–3.

Table 124–2—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1 register	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable register	1.9.0	PMD_global_transmit_disable
PMD transmit disable 3 to PMD transmit disable 0	PMD transmit disable register	1.9.4 to 1.9.1	PMD_transmit_disable_3 to PMD_transmit_disable_0

Table 124–3—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1 register	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2 register	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2 register	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect register	1.10.0	PMD_global_signal_detect
PMD receive signal detect 3 to PMD receive signal detect 0	PMD receive signal detect register	1.10.4 to 1.10.1	PMD_signal_detect_3 to PMD_signal_detect_0

124.5 PMD functional specifications

The 400GBASE-DR4 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

124.5.1 PMD block diagram

The PMD block diagram is shown in Figure 124–2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 124.8 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 124.11.3). Unless specified otherwise, all receiver measurements and tests defined in 124.8 are made at TP3.

TP1<0:3> and TP4<0:3> are optional reference points that may be useful to implementors for testing components (these test points will not typically be accessible in an implemented system).

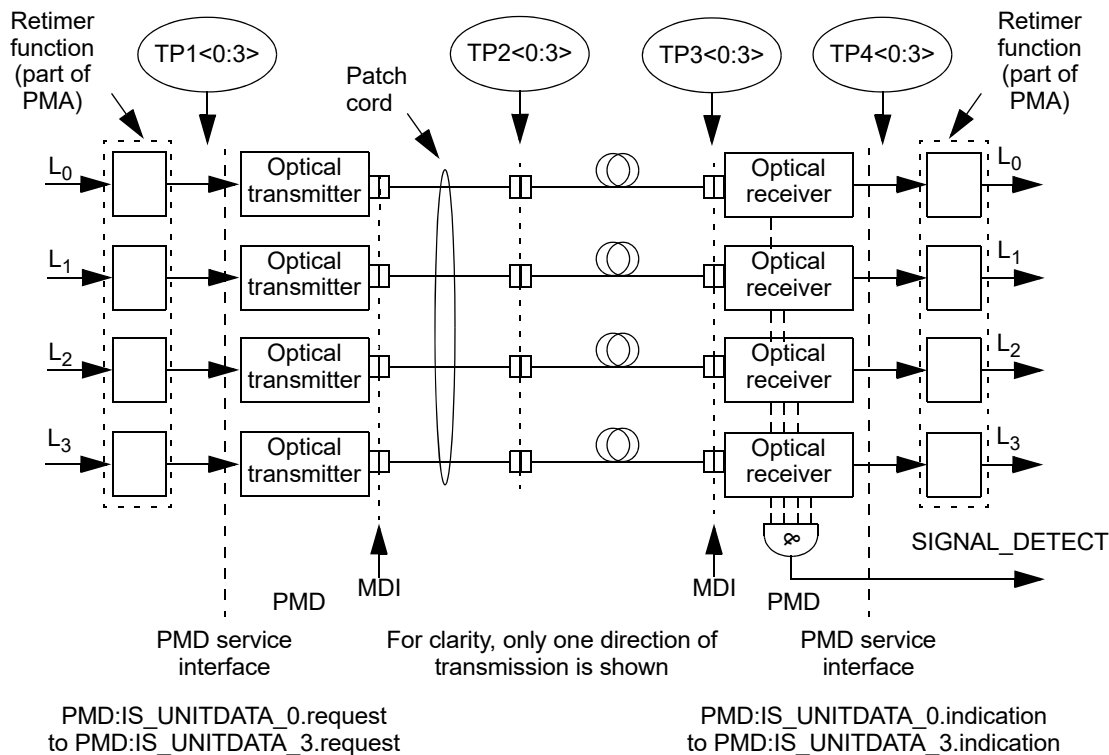


Figure 124-2—Block diagram for 400GBASE-DR4 transmit/receive paths

124.5.2 PMD transmit function

The PMD Transmit function shall convert the four symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_0.request to PMD:IS_UNITDATA_3.request into four separate optical signals. The four optical signals shall then be delivered to the MDI, which contains four parallel light paths for transmit, according to the transmit optical specifications in this clause. The highest optical power level in each signal shall correspond to tx_symbol = three and the lowest shall correspond to tx_symbol = zero.

124.5.3 PMD receive function

The PMD Receive function shall convert the four parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_3.indication, all according to the receive optical specifications in this clause. The higher optical power level in each signal shall correspond to rx_symbol = three and the lowest shall correspond to rx_symbol = zero.

124.5.4 PMD global signal detect function

The PMD global signal detect function shall report the state of SIGNAL_DETECT via the PMD service interface. The SIGNAL_DETECT parameter is signaled continuously, while the PMD:IS_SIGNAL.indication message is generated when a change in the value of SIGNAL_DETECT occurs. The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the inter-sublayer service interface primitives defined in 116.3.

SIGNAL_DETECT shall be a global indicator of the presence of optical signals on all four lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 124-4.

The PMD receiver is not required to verify whether a compliant 400GBASE-DR4 signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Table 124–4—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ -15 dBm	FAIL
For all lanes; [(Optical power at TP3 $\geq$ average receive power, each lane (min) Table 124–7) AND (compliant 400GBASE-R signal input)]	OK
All other conditions	Unspecified

As a consequence of the requirements for the setting of the SIGNAL_DETECT parameter, implementations need to provide adequate margin between the input optical power level at which the SIGNAL_DETECT parameter is set to OK, and the inherent noise level of the PMD including the effects of crosstalk, power supply noise, etc.

Various implementations of the Signal Detect function are permitted by this standard, including implementations that generate the SIGNAL_DETECT parameter values in response to the amplitude of the modulation of the optical signal and implementations that respond to the average optical power of the modulated optical signal.

124.5.5 PMD lane-by-lane signal detect function

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each PMD_signal_detect_*i*, where *i* represents the lane number in the range 0:3, shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 124–4.

124.5.6 PMD reset function

If the MDIO interface is implemented, and if PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

124.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

- When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 124–6.
- If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

124.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitters in each lane to be selectively disabled.

- a) When a `PMD_transmit_disable_i` variable (where i represents the lane number in the range 0:3) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 124–6.
- b) If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disable_i` variable to one, turning off the optical transmitter in each lane.

If the optional PMD lane-by-lane transmit disable function is not implemented in MDIO, an alternative method may be provided to independently disable each transmit lane for testing purposes.

124.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set `PMD_fault` to one.

If the MDIO interface is implemented, `PMD_fault` shall be mapped to the fault bit as specified in 45.2.1.2.3.

124.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the `PMD_transmit_fault` variable to one.

If the MDIO interface is implemented, `PMD_transmit_fault` shall be mapped to the transmit fault bit as specified in 45.2.1.7.4.

124.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the `PMD_receive_fault` variable to one.

If the MDIO interface is implemented, `PMD_receive_fault` shall be mapped to the receive fault bit as specified in 45.2.1.7.5.

124.6 Lane assignments

There are no lane assignments (within a group of transmit or receive lanes) for 400GBASE-DR4. While it is expected that a PMD will map electrical lane i to optical lane i and vice versa, there is no need to define the physical ordering of the lanes, as the PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in 124.11.3.1.

124.7 PMD to MDI optical specifications for 400GBASE-DR4

The operating range for the 400GBASE-DR4 PMD is defined in Table 124–5. A 400GBASE-DR4 compliant PMD operates on type B1.1, B1.3, or B6_a single-mode fibers according to the specifications defined in Table 124–12. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-DR4 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Table 124–5—400GBASE-DR4 operating range

PMD type	Required operating range
400GBASE-DR4	2 m to 500 m

124.7.1 400GBASE-DR4 transmitter optical specifications

The 400GBASE-DR4 transmitter shall meet the specifications defined in Table 124–6 per the definitions in 124.8.

Table 124–6—400GBASE-DR4 transmit characteristics

Description	Value	Unit
Signaling rate, each lane (range)	53.125 ± 100 ppm	GBd
Modulation format	PAM4	—
Lane wavelength (range)	1304.5 to 1317.5	nm
Side-mode suppression ratio (SMSR), (min)	30	dB
Average launch power, each lane (max)	4	dBm
Average launch power, each lane ^a (min)	–2.9	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.2	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) ^b	–0.8	dBm
Launch power in OMA _{outer} minus TDECQ, each lane (min)	–2.2	dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4	dB
TDECQ – $10\log_{10}(C_{eq})^c$ (max)	3.4	dB
Average launch power of OFF transmitter, each lane (max)	–15	dBm
Extinction ratio, each lane (min)	3.5	dB
Transmitter transition time (max)	17	ps
RIN _{21.4} OMA (max)	–136	dB/Hz
Optical return loss tolerance (max)	21.4	dB
Transmitter reflectance ^d (max)	–26	dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b Even if the TDECQ < 1.4 dB, the OMA_{outer} (min) has to exceed this value.

^c C_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

^d Transmitter reflectance is defined looking into the transmitter.

124.7.2 400GBASE-DR4 receive optical specifications

The 400GBASE-DR4 receiver shall meet the specifications defined in Table 124–7 per the definitions in 124.8. See NOTE at the end of 120.5.2 concerning the transition density of lanes operating at this nominal signaling rate.

Table 124–7—400GBASE-DR4 receive characteristics

Description	Value	Unit
Signaling rate, each lane (range)	53.125 ± 100 ppm	GBd
Modulation format	PAM4	—
Lane wavelengths (range)	1304.5 to 1317.5	nm
Damage threshold ^a , each lane	5	dBm
Average receive power, each lane (max)	4	dBm
Average receive power, each lane ^b (min)	–5.9	dBm
Receive power (OMA _{outer}), each lane (max)	4.2	dBm
Receiver reflectance (max)	–26	dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max)	Equation (124–1)	dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^d (max)	–1.9	dBm
Conditions of stressed receiver sensitivity test: ^e		
Stressed eye closure for PAM4 (SECQ), lane under test	3.4	dB
SECQ – $10\log_{10}(C_{eq})^f$ (max), lane under test	3.4	dB
OMA _{outer} of each aggressor lane	4.2	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Receiver sensitivity (OMA_{outer}), each lane (max) is optional and is defined for a transmitter with a value of SECQ up to 3.4 dB.

^d Measured with conformance test signal at TP3 (see 124.8.10) for the BER specified in 124.1.1.

^e These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^f C_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

124.7.3 400GBASE-DR4 illustrative link power budget

An illustrative power budget and penalties for 400GBASE-DR4 channels are shown in Table 124–8.

Table 124–8—400GBASE-DR4 illustrative link power budget

Parameter	Value	Unit
Power budget (for max TDECQ)	6.5	dB
Operating distance	500	m
Channel insertion loss ^a	3	dB
Maximum discrete reflectance	See 124.11.2.2	dB
Allocation for penalties ^b (for max TDECQ)	3.5	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 124–5 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 124.11.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

124.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

124.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 124–10 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 124–10 may be used to perform that test. The test patterns used in this clause are shown in Table 124–9.

Table 124–9—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.4
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle	119.2.4.9
6	SSPRQ	120.5.11.2.3

Table 124–10—Test-pattern definitions and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6 or valid 400GBASE-R signal	124.8.2
Side mode suppression ratio	3, 5, 6 or valid 400GBASE-R signal	124.8.2
Average optical power	3, 5, 6 or valid 400GBASE-R signal	124.8.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	124.8.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	124.8.5
Extinction ratio	4 or 6	124.8.6
Transmitter transition time	Square wave or 6	124.8.7
RIN _{21.4OMA}	Square wave	124.8.8
Stressed receiver conformance test signal calibration	6	124.8.10
Stressed receiver sensitivity	3 or 5	124.8.10

124.8.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the range given in Table 124–6 if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 124–10.

124.8.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 124–6 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 124–10.

124.8.4 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 124–6. The OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 124–10 as the difference between the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 124–3.

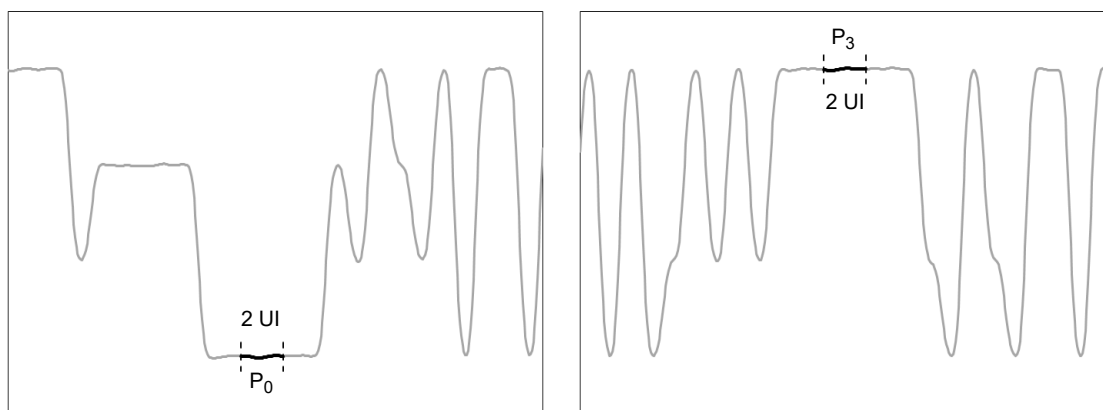


Figure 124–3—Example power levels P_0 and P_3 from PRBS13Q test pattern

124.8.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ and $TDECQ - 10\log_{10}(C_{eq})$ of each lane shall be within the limits given in Table 124–6 if measured using the methods specified in 121.8.5.1, 121.8.5.2, and 121.8.5.3 using a reference equalizer as described in 121.8.5.4 where T is the symbol period for 400GBASE-DR4, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 124–6 and uses the test pattern specified for TDECQ in Table 124–10.
- The combination of the O/E converter and the oscilloscope has a 3 dB bandwidth of approximately 26.5625 GHz with a fourth-order Bessel-Thomson response to at least 1.3×53.125 GHz, and at frequencies above 1.3×53.125 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The normalized noise power density spectrum $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 26.5625 GHz.

124.8.6 Extinction ratio

The extinction ratio of each lane shall be within the limits given in Table 124–6 if measured using a test pattern specified for extinction ratio in Table 124–10. The extinction ratio of a PAM4 optical signal is

defined as the ratio of the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 124–3.

124.8.7 Transmitter transition time

The transmitter transition time of each lane shall be within the limits given in Table 124–6 if measured using a test pattern specified for transmitter transition time in Table 124–10.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} , or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively, as measured through an O/E converter and oscilloscope with response defined as follows. The combined response of the O/E converter and oscilloscope has a 3 dB bandwidth of approximately 26.5625 GHz with a fourth-order Bessel-Thomson response to at least 1.3×53.125 GHz. At frequencies above 1.3×53.125 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 124.8.4), with the exception that the square wave test pattern can be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

124.8.8 Relative intensity noise ($RIN_{21.4OMA}$)

RIN shall be as defined by the measurement methodology of 52.9.6 with the following exceptions:

- The optical return loss is 21.4 dB.
- Each lane may be tested individually with the sum of the optical power from all of the lanes not under test being below –30 dBm.
- The upper –3 dB limit of the measurement apparatus is to be approximately equal to the signaling rate (i.e., 53.2 GHz).
- The test pattern is according to Table 124–10.

124.8.9 Receiver sensitivity

Receiver sensitivity is optional and is defined for a transmitter with a value of $SECQ$ up to 3.4 dB. Receiver sensitivity should meet Equation (124–1), which is illustrated in Figure 124–4.

$$RS = \max(-3.9, SECQ - 5.3) \quad (\text{dBm}) \quad (124-1)$$

where

RS is the receiver sensitivity
 $SECQ$ is the $SECQ$ of the transmitter used to measure the receiver sensitivity

The normative requirement for receivers is stressed receiver sensitivity.

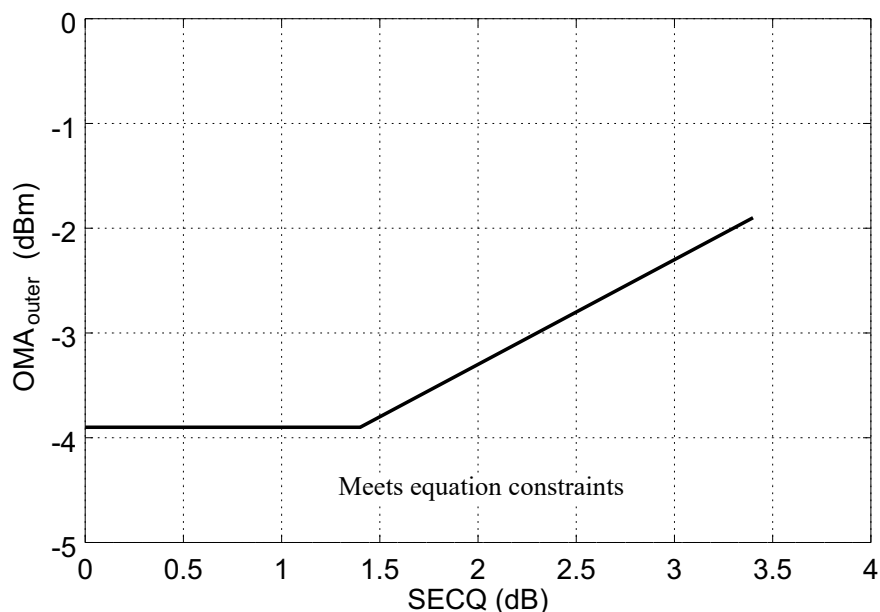


Figure 124-4—Illustration of receiver sensitivity

124.8.10 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 124-7 if measured using the method defined in 121.8.10 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 124.8.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 124-6.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $RIN_{21.4OMA}$ of the SRS test source should be no greater than the value specified in Table 124-6.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 124-6 using test patterns specified in Table 124-10.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test”, “ $SECQ - 10\log_{10}(C_{eq})$ (max), lane under test”, and “ OMA_{outer} of each aggressor lane” are as given in Table 124-7.

124.9 Safety, installation, environment, and labeling

124.9.1 General safety

All equipment subject to this clause shall conform to IEC 60950-1.

124.9.2 Laser safety

400GBASE-DR4 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.²¹⁴

124.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

124.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 400GBASE-DR4 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

²¹⁴A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

124.9.5 Electromagnetic emission

A system integrating a 400GBASE-DR4 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

124.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

124.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 400GBASE-DR4).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 124.9.2.

124.10 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 124–5.

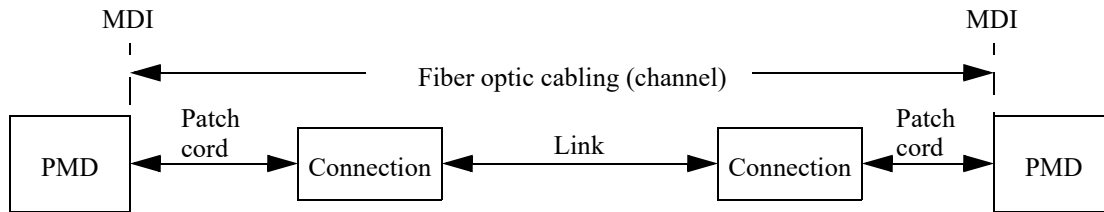


Figure 124–5—Fiber optic cabling model

The channel insertion loss is given in Table 124–11. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term channel is used here for consistency with generic cabling standards.

124.11 Characteristics of the fiber optic cabling (channel)

The 400GBASE-DR4 fiber optic cabling shall meet the specifications defined in Table 124–12. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

124.11.1 Optical fiber cable

The fiber optic cable requirements are satisfied by cables containing IEC 60793-2-50 type B1.1 (dispersion un-shifted single-mode), type B1.3 (low water peak single-mode), or type B6_a (bend insensitive) fibers or the requirements in Table 124–12 where they differ.

Table 124–11—Fiber optic cabling (channel) characteristics

Description	400GBASE-DR4	Unit
Operating distance (max)	500	m
Channel insertion loss ^{a,b} (max)	3	dB
Channel insertion loss (min)	0	dB
Positive dispersion ^b (max)	0.8	ps/nm
Negative dispersion ^b (min)	–0.93	ps/nm
DGD_max ^c	2.24	ps
Optical return loss (min)	37	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b Over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

Table 124–12—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.093	ps/nm ² km

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3.

124.11.2 Optical fiber connection

An optical fiber connection, as shown in Figure 124–5, consists of a mated pair of optical connectors.

124.11.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 2.75 dB total connection and splice loss. For example, this allocation supports five connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 124–11 are met.

124.11.2.2 Maximum discrete reflectance

The maximum value for each discrete reflectance shall be less than or equal to the value shown in Table 124–13 corresponding to the number of discrete reflectances above –55 dB within the channel. For numbers of discrete reflectances in between two numbers shown in the table, the lower of the two corresponding maximum discrete reflectance values applies.

Table 124–13—Maximum value of each discrete reflectance

Number of discrete reflectances above –55 dB	Maximum value for each discrete reflectance
1	–37 dB
2	–42 dB
4	–45 dB
6	–47 dB
8	–48 dB
10	–49 dB

124.11.3 Medium Dependent Interface (MDI)

The 400GBASE-DR4 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 124–5). The 400GBASE-DR4 PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 124–7. Example constructions of the MDI include the following:

- a) PMD with a connectorized fiber pigtail plugged into an adapter
- b) PMD receptacle

124.11.3.1 Optical lane assignments

The four transmit and four receive optical lanes of 400GBASE-DR4 shall occupy the positions depicted in Figure 124–6 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains eight active lanes within twelve total positions. The transmit optical lanes occupy the left-most four positions. The receive optical lanes occupy the right-most four positions. The four center positions are unused.

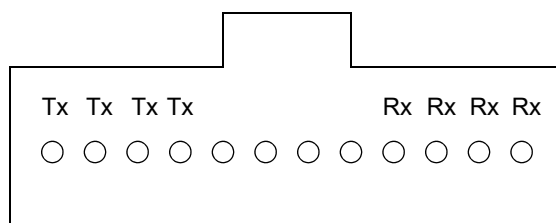


Figure 124–6—400GBASE-DR4 optical lane assignments

124.11.3.2 Medium Dependent Interface (MDI) requirements

The MDI shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-9: *MPO device receptacle, angled interface*. The plug terminating the optical fiber cabling shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-1: *MPO female plug connector, down-angled interface for 2 to 12 fibres*. The MDI shall optically mate with the plug on the optical fiber cabling. Figure 124–7 shows an MPO female plug connector with down-angled interface, and an MDI as an active device receptacle with angled interface.

The MDI shall meet the interface performance specifications of IEC 61753-021-2 for performance level D/2.

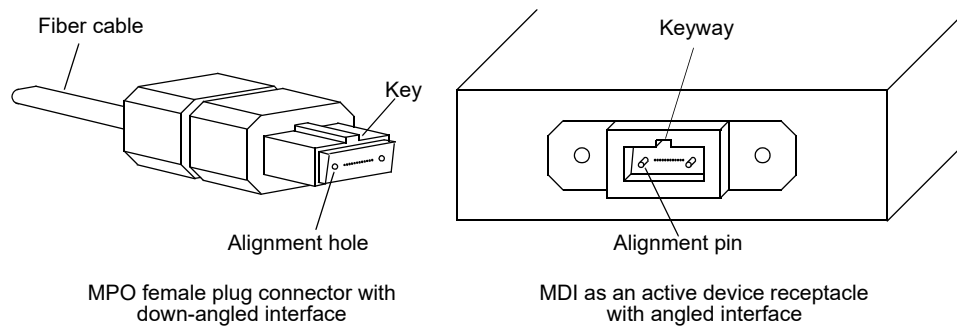


Figure 124-7—MPO female plug with down-angled interface and MDI active device receptacle with angled interface

NOTE—Transmitter compliance testing is performed at TP2 as defined in 124.5.1, not at the MDI.

124.12 Protocol implementation conformance statement (PICS) proforma for Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4²¹⁵

124.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

124.12.2 Identification

124.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

124.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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²¹⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

124.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*INS	Installation / cable	124.10	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	124.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	124.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	124.3.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	124.3.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	124.4	Registers and interface supported	O	Yes [] No []

124.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 40GBASE-DR4

124.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 40GBASE-R PCS and PMA	124.1		M	Yes []
F2	Integration with management functions	124.1		O	Yes [] No []
F3	Bit error ratio	124.1.1	Meets the BER specified in 124.1.1	M	Yes []
F4	Transmit function	124.5.2	Conveys symbols from PMD service interface to MDI	M	Yes []
F5	Mapping between optical signal and logical signal for transmitter	124.5.2	Highest optical power corresponds to tx_symbol = three	M	Yes []
F6	Receive function	124.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F7	Conversion of four optical signals to four electrical signals	124.5.3	For delivery to the PMD service interface	M	Yes []
F8	Mapping between optical signal and logical signal for receiver	124.5.3	Highest optical power corresponds to rx_symbol = three	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F9	Global Signal Detect function	124.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_DETECT)	M	Yes []
F10	Global Signal Detect behavior	124.5.4	SIGNAL_DETECT is a global indicator of the presence of optical signals on all four lanes	M	Yes []
F11	Lane-by-lane Signal Detect function	124.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 124–4	MD:O	Yes [] No [] N/A []
F12	PMD reset function	124.5.6	Resets the PMD sublayer	MD:O	Yes [] No [] N/A []

124.12.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	124.4		MD:M	Yes [] N/A []
M2	Global transmit disable function	124.5.7	Disables all of the optical transmitters with the PMD_global_transmit_disable variable	MD:O	Yes [] No [] N/A []
M3	PMD lane-by-lane transmit disable function	124.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	MD:O	Yes [] No [] N/A []
M4	PMD_fault function	124.5.9	Sets PMD_fault to one if any local fault is detected	MD:O	Yes [] No [] N/A []
M5	PMD_transmit_fault function	124.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	MD:O	Yes [] No [] N/A []
M6	PMD_receive_fault function	124.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	MD:O	Yes [] No [] N/A []

124.12.4.3 PMD to MDI optical specifications for 400GBASE-DR4

Item	Feature	Subclause	Value/Comment	Status	Support
DR1	Transmitter meets specifications in Table 124–6	124.7.1	Per definitions in 124.8	M	Yes []
DR2	Receiver meets specifications in Table 124–7	124.7.2	Per definitions in 124.8	M	Yes []

124.12.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	124.8	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	124.8.2	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	124.8.3	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	124.8.4	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	124.8.5	Each lane	M	Yes []
OM6	Extinction ratio	124.8.6	Each lane	M	Yes []
OM7	Transmitter transition time	124.8.7	Each lane	M	Yes []
OM8	Stressed receiver sensitivity	124.8.10	Each lane	M	Yes []

124.12.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	124.9.1	Conforms to IEC 60950-1	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	124.9.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	124.9.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

124.12.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	124.11	Meets requirements specified in Table 124–11	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	124.11.2.2	Meets requirements specified in Table 124–13	INS:M	Yes [] N/A []
OC3	MDI layout	124.11.3.1	Optical lane assignments per Figure 124–6	M	Yes []
OC4	MDI dimensions	124.11.3.2	Per IEC 61754-7-1 interface 7-1-9	M	Yes []
OC5	Cabling connector dimensions	124.11.3.2	Per IEC 61754-7-1 interface 7-1-1	INS:M	Yes [] N/A []
OC6	MDI mating	124.11.3.2	MDI optically mates with plug on the cabling	M	Yes []